

Lecture 4

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Homework

Lets go over it!!



Last week

- ◆ 2 Small Papers
- ◆ Software dev life cycle
- ◆ Programming paradigms

Functions

Math



Combinatorics

- ◆ Function notation
- ◆ Counting for adults...

What is a function?

- ◆ Portable series of instructions.
- ◆ $f(x, y) = x + y \longrightarrow f \quad x \quad y = x + y$

Defining functions

- ◆ Name: bind the Logic to a callable name
- ◆ Type: the logic should have an output
- ◆ Parameters: function inputs (arguments)
- ◆ Logic: define function logic

“Simplicity is a great virtue but it requires hard work to achieve it and education to appreciate it. And to make matters worse: complexity sells better.”

-Edsger W. Dijkstra

Build examples

- ◆ Sum function
- ◆ Summation function
- ◆ Euclidian distance function

Sum function

sum : *Number* $\rightarrow$ *Number* $\rightarrow$ *Number*

$$\textit{sum}(x, y) = x + y$$

Parameter list



The diagram illustrates the components of the function definition. A red curly brace under the parameter list (x, y) has a red arrow pointing down to the text 'Parameter list'. Another red curly brace under the logic $x + y$ has a red arrow pointing down to the text 'Logic'.

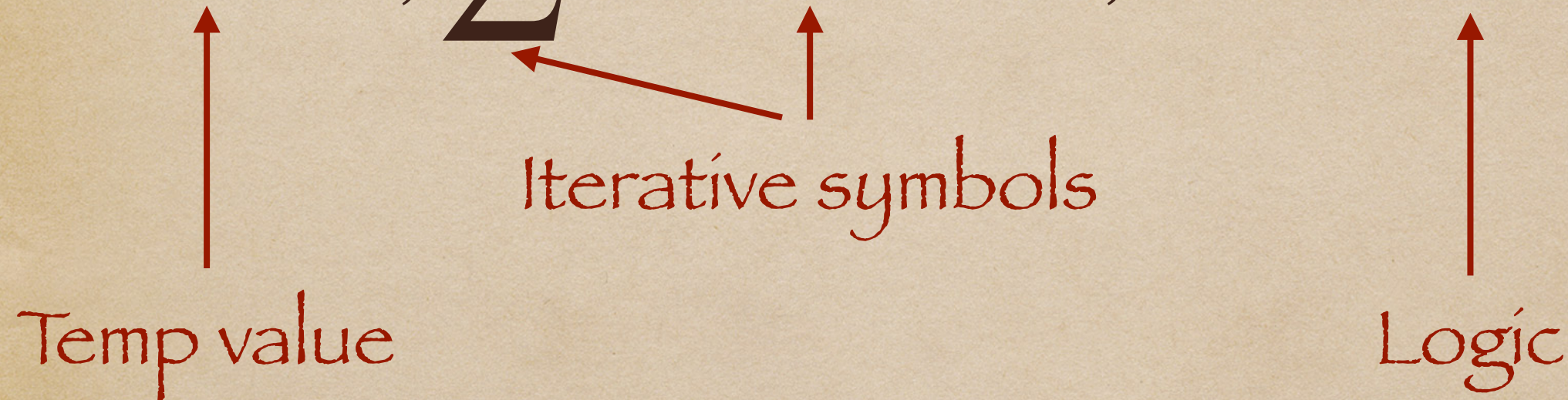
Logic

Summation function

$\Sigma : \textit{Collection} \rightarrow \textit{Number}$

$sum \leftarrow 0, \Sigma A = \forall x \in A, sum \leftarrow x + sum$

Temp value



Iterative symbols

Logic

Euclidian distance

$$\alpha = (x, y)$$

$$\Delta : \alpha \rightarrow \alpha \rightarrow \textit{Number}$$

$$\Delta p_0 p_1 = c, \text{ where}$$

$$a \leftarrow |x_1 - x_2|$$

$$b \leftarrow |y_1 - y_0|$$

$$c \leftarrow \sqrt{a_2 + b_2}$$

“Computer Science is no more about computers
than astronomy is about telescopes”

-Edsger W. Dijkstra

Iteration



How do we use Iteration

- ◆ Pseudocode: $\forall, \Sigma, \Pi \dots$
- ◆ Python: *for, while, loop, recursion*

Live Coding